

● MEDIUM

● **ADVANCED MATH**

Advanced Math

01

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1

GRID-IN ADVANCED MATH

The function f is defined by $fx = 5\frac{1}{4} \cdot x^2 + \frac{11}{4}$. What is the value of $f\frac{1}{4}$?

STUDENT-PRODUCED RESPONSE

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.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Which quadratic equation has exactly one distinct real solution?

A. $x + 15^2 = 0$

B. $x + 15^2 = -45$

C. $x + 15^2 = 45$

D. $x + 15^2 = 135$

If $x^2 = a + b$ and $y^2 = a + c$, which of the following is equal to $(x^2 - y^2)^2$?

A. $a^2 - 2ac + c^2$

B. $b^2 - 2bc + c^2$

C. $4a^2 - 4abc + c^2$

D. $4a^2 - 2abc + b^2c^2$

A physics class is planning an experiment about a toy rocket. The equation $y = -16x^2 - 5.6x + 502$ gives the estimated height y , in feet, of the toy rocket x seconds after it is launched into the air. Which of the following is the best interpretation of the vertex of the graph of the equation in the xy -plane?

- A. This toy rocket reaches an estimated maximum height of 502 feet 16 seconds after it is launched into the air.
- B. This toy rocket reaches an estimated maximum height of 502 feet 5.6 seconds after it is launched into the air.
- C. This toy rocket reaches an estimated maximum height of 16 feet 502 seconds after it is launched into the air.
- D. This toy rocket reaches an estimated maximum height of 5.6 feet 502 seconds after it is launched into the air.

$$p = \frac{k}{4j + 9}$$

The given equation relates the distinct positive numbers p , k , and j . Which equation correctly expresses $4j + 9$ in terms of p and k ?

- A. $4j + 9 = \frac{k}{p}$
- B. $4j + 9 = kp$
- C. $4j + 9 = k - p$
- D. $4j + 9 = \frac{p}{k}$

$$f(x) = x^2 + bx$$

$$g(x) = 9x^2 - 27x$$

Functions f and g are given, and in function f , b is a constant. If

$f(x) \cdot g(x) = 9x^4 - 26x^3 - 3x^2$, what is the value of b ?

A. -26

B. $-\frac{26}{9}$

C. $\frac{1}{9}$

D. 9

$$f(x) = 4,000(0.75)^x$$

An entomologist recommended a program to reduce a certain invasive beetle population in an area. The given function estimates this beetle species' population x years after 2012, where $x \leq 7$. Which of the following is the best interpretation of 4,000 in this context?

- A. The estimated initial beetle population for this species and area in 2012
- B. The estimated beetle population for this species and area 7 years after 2012
- C. The estimated percent decrease in the beetle population for this species and area each year after 2012
- D. The estimated percent decrease in the beetle population for this species and area every 7 years after 2012

Q8

GRID-IN

ADVANCED MATH

$$y = x^2 - 4x + 4$$
$$y = 4 - x$$

If the ordered pair (x, y) satisfies the system of equations above, what is one possible value of x ?

STUDENT-PRODUCED RESPONSE

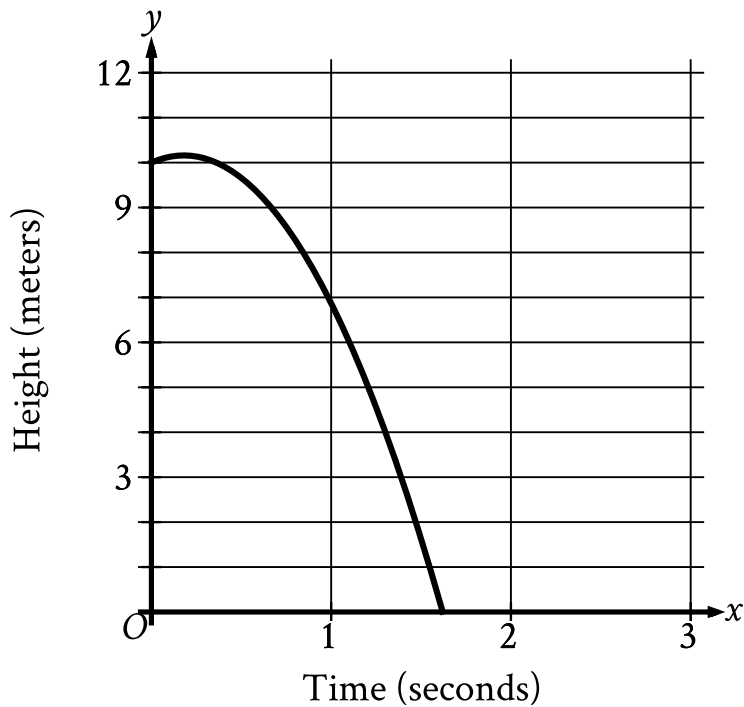
.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$4a^2 + 20ab + 25b^2$$

Which of the following is a factor of the polynomial above?

- A. $a + b$
- B. $2a + 5b$
- C. $4a + 5b$
- D. $4a + 25b$



- The parabola opens downward.
- The vertex is at the approximate point (0.2 comma 10.2).
- The parabola passes through the following points:
 - (0 comma 10)
 - approximately (0.2 comma 10.2)
 - approximately (1 comma 6.9)
 - approximately (1.6 comma 0)

A competitive diver dives from a platform into the water. The graph shown gives the height above the water y , in meters, of the diver x seconds after diving from the platform. What is the best interpretation of the x -intercept of the graph?

- A. The diver reaches a maximum height above the water at 1.6 seconds.
- B. The diver hits the water at 1.6 seconds.

- C. The diver reaches a maximum height above the water at 0.2 seconds.
- D. The diver hits the water at 0.2 seconds.